

Pictures Become Numbers

Project: Helpful Classroom Sorter

1. Pictures Become Numbers

Teach the learning cycle

Primary target: Encode and reconstruct a small image using an explicit grid convention

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Look for: Execute a stated rule including its boundary

Reference example, not the only acceptable wording: At least 5 gives narrow, wide, wide. Greater than 5 gives narrow, narrow, wide. Only the equality case changes among these inputs.

Input, Process, Output

Without opening the old lesson, explain one key idea from Input, Process, Output. Give an example and a mistake that the idea helps you avoid.

Look for: Trace information through a system without confusing information with a physical action

Reference example, not the only acceptable wording: Input includes both request fields and all three catalogue records. The process checks science AND pages $\leq$ 120. Output contains A only; B fails the length condition and C fails the genre condition. Listing A does not establish that a copy was reserved.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

Concept to explain

A digital image is a grid of measured color values.

Term · Meaning

Term: pixel

Meaning: one picture element in a grid

Term: grid

Meaning: rows and columns that locate values

Term: channel

Meaning: one measured color component

Term: resolution

Meaning: number of pixels used to represent an image

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

The image is binary for transparency, not a claim that ordinary colour images contain only zeros and ones as pixel values. Distinguish bit-level storage from the image value convention. Grid reconstruction is assessed separately from interpretation. A learner may name the worked pattern differently while still reconstructing it correctly.

Readiness answer

No. The fitting examples may not distinguish the rules, and agreement on those examples does not settle every unseen case.

Squared paper and a dark pencil for each learner

The worked grid with its explicit reading convention

Two empty two-row four-column grids for independent assessment

End goal for the pictorial case

By the end, I can encode pixel positions and show why a count cannot reconstruct a picture.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT light measured at grid locations

PROCESS store and transform pixel values

OUTPUT a displayed, resized, blurred, or classified image

CORE IDEA A digital image is a grid of measured color values.

UNDER THE HOOD Digital images sample a scene across a spatial grid. Each pixel stores channel values, commonly red, green, and blue. Resolution controls sampling density; transformations operate on numbers, not on an intact object inside the computer.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each pixel position.

STEP 1 — OBSERVE light measured at grid locations

STEP 2 — TRACE store and transform pixel values

STEP 3 — CONCLUDE a displayed, resized, blurred, or classified image

MODEL REASONING Each grid position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions. The computer transforms numbers; the human viewer recognizes the shape.

DO NOT OVERCLAIM A pixel is a measurement at a location, not a tiny copy of the object.

What new evidence could change this answer?

Concrete teaching case

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Exact explanation to model

Initially top-right and bottom-left are black. The change produces three black squares. Pixel values encode an image; object recognition is a separate interpretation.

Explain the added worked example

Imagine sending a small black-and-white sign using only numbers. We can divide the sign into a grid, record 1 for a black square and 0 for a white square, and read each row from left to right, starting at the top. The receiver needs the width, height and these conventions to reconstruct the same arrangement.

A pixel is a position with a recorded value. In our simplified binary image, one value selects black or white. Other image formats can record brightness levels or multiple colour channels. The convention 1 means black belongs to this exercise; it is not a universal rule for every digital image.

Encoding a picture does not recognise what it depicts. A program can store or display the numbers without calling the image a letter, an arrow or a person. Interpretation is another operation, just as displaying a

book title was different from reading the entire book in Week 2.

The first row is 0,1,0, so only its middle pixel is black. The second row is 1,1,1, so all three are black. The third row repeats the first. This reconstructs the displayed cross-like pattern under our convention.

A flattened record would be 0,1,0,1,1,1,0,1,0. Without the grid dimensions, the same nine values could also be drawn in one row of nine. Keep the shape information with the values. Width times height counts pixels; it does not tell us their contents.

Key terms

Term · Meaning

Term: Pixel

Meaning: One position in a digital image grid with a recorded value or values

Term: Encoding

Meaning: A convention for representing information with symbols or numbers

Term: Resolution

Meaning: The number of positions in an image grid, often given as width by height

Term: Spatial arrangement

Meaning: Where elements are located relative to one another

Term: Interpretation

Meaning: A meaning or category assigned to a representation using a rule, model or human judgment

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Suppose we discard the row records and retain only black pixel count=5. That count matches the cross, but it also matches rows 1,1,1 then 1,1,0 then 0,0,0. The second arrangement fills the top row and two positions below it. Both have five black pixels and nine pixels in total, yet they are different images.

Width 3 and height 3 · There are 9 positions · The value at each position

Width 3 and height 3: Black pixel count 5

There are 9 positions: Five positions are black under this code

The value at each position: Which five positions

Width 3 and height 3: Dimensions and all three ordered rows

There are 9 positions: The full binary grid under this convention

The value at each position: What a viewer or model will call the picture

A pixel count can still be a useful feature for some tasks. For example, it can separate empty grids from grids with at least one mark. But it cannot separate every shape. To judge a feature, ask whether it retains the distinctions required by the task.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Pictures Become Numbers

Two fictional 2-by-3 pixel badges contain the same number of dark squares.

The facts we will use

Dark=1, light=0; rows are read top to bottom, left to right.

Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Pixel convention

Each image has two rows and three columns; 1 means dark.



2. Badge A

Top 101, bottom 010: three dark pixels.



3. Badge B

Top 111, bottom 000: also three dark pixels.



4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Pixel convention

What happens and why: Each image has two rows and three columns; 1 means dark.

Step: 2. Badge A

What happens and why: Top 101, bottom 010: three dark pixels.

Step: 3. Badge B

What happens and why: Top 111, bottom 000: also three dark pixels.

Step: 4. Compare arrangements

What happens and why: The totals agree, but middle-top and middle-bottom positions differ.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A grid preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

What this example does not establish

This toy binary encoding does not include real photographs' colour channels, compression or illumination effects.

Change one thing in the pictured case

Change: Change A's bottom-left pixel from 0 to 1.

Prediction: A now has four dark pixels; its row encoding is 101 / 110.

Reason: Exactly one light pixel becomes dark; the other five positions remain unchanged.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Write the question to explain: Draw it, then change the top-left value to 1.

Reveal after predictions: Initially top-right and bottom-left are black. The change produces three black squares. Pixel values encode an image; object recognition is a separate interpretation.

Keep for an independent attempt: Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left pixel to light. Which number changes?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The two rows are black-white-black and white-black-white, giving three black pixels. Other arrangements share that count, so it cannot recover positions. For the exit, move one black pixel to a previously white position while preserving the total; the image changes but the count does not.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Recall the tied-threshold example. Ask why indistinguishable evidence can leave different answers possible.

Stage: 7 to 20 minutes

Teacher action: Agree on colour and reading conventions. Reconstruct the worked grid row by row.

Stage: 20 to 33 minutes

Teacher action: Flatten the values and show why dimensions matter. Compare the two five-black-pixel patterns.

Stage: 33 to 45 minutes

Teacher action: Pairs complete W1 and W2, exchanging encodings and reading them back.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Show two equal-count grids and ask which input a count-only classifier would receive.

Stage: 8 to 23 minutes

Teacher action: Complete W3 and diagnose a reversed convention. Require a specific counterexample.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently. Check the dimensions and counts before evaluating the explanation.

Stage: 38 to 45 minutes

Teacher action: Discuss W5 and the exit question. Distinguish stored values from assigned meaning.

Reduce help deliberately

Completion task: A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

Teacher check: One row of width 6 is [1,0,0,1,1,0]. Two rows of width 3 are [1,0,0] and [1,1,0]. Specify width and height, plus reading order and colour convention. The values alone do not choose the shape.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Images as numerical grids. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Our convention states that meaning. Other encodings use different scales or channels; read their specification.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Find a position where the original badges differ, then read the changed A back as a picture.

Reasoned teacher answer

The top-middle pixel is 0 in A and 1 in B. Changed A is dark at top-left, top-right, bottom-left and bottom-middle.

Changed-case reasoning: A now has four dark pixels; its row encoding is 101 / 110. Exactly one light pixel becomes dark; the other five positions remain unchanged.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Decode a tiny pixel picture and predict a blurred version.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR An image is stored as its object name.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The number one always means black: Our convention states that meaning. Other encodings use different scales or channels; read their specification.

A pixel count is the picture: Have students reconstruct two different arrangements with the same count.

Storing pixels recognises the object: A grid decoder only reconstructs the representation. A separate interpretation step assigns a category.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Initially top-right and bottom-left are black. The change produces three black squares. Pixel values encode an image; object recognition is a separate interpretation.

Independent question

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left pixel to light. Which number changes?

Expected independent answer

The original light pixels are top-left and bottom-right. The bottom-left entry changes from 0 to 1, giving rows [1,0] and [1,1]. Position and value both matter.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “An image is stored as its object name.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Row-major encoding is [1,0,1,0,1,0], with shape 2×3. Interpreting the same list column-first gives top [1,1,1] and bottom [0,0,0]. Values need shape and ordering conventions; encoding alone does not recognise an object.

Explanations for the additional workbook examples

W1 Read and reconstruct

Using 1 for black and 0 for white, draw width 3, height 2 with rows 1,0,1 and 0,1,0. Read it back into numbers and count black pixels.

The upper row has black, white, black; the lower row has white, black, white. Reading back gives the supplied two rows. There are three black pixels among six total positions.

W2 Preserve the missing field

A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

One row of width 6 is [1,0,0,1,1,0]. Two rows of width 3 are [1,0,0] and [1,1,0]. Specify width and height, plus reading order and colour convention. The values alone do not choose the shape.

W3 Test a count feature

A classifier receives only the number of black pixels. Can it always distinguish the chapter's cross from every other three by three pattern? Construct a different pattern with the same count and explain.

No. Rows [1,1,1], [1,1,0], [0,0,0] have five black pixels, as does the cross, but a different arrangement. The classifier receives identical input counts for the two, so that feature alone cannot distinguish them.

W4 Design a new counterexample

Create two different images of width 4 and height 2, each with exactly three black pixels. Write both complete row encodings. Explain which fact their shared count retains and which difference it discards.

One valid pair is A [1,1,1,0]/[0,0,0,0] and B [1,0,0,0]/[0,1,0,1]. Both have two rows of four values and three ones. Count retains the number of black positions but discards their locations. Other valid pairs are accepted if dimensions, counts and arrangements meet the constraints.

W5 Locate an input problem

A program receives the correct grid but displays black wherever the code says 0 and white wherever it says 1. A student proposes collecting more classifier examples. Identify the earlier mismatch and the first check to perform.

The display uses the opposite colour convention from the sender. Check and align the encoding/decoding convention, then reconstruct a known grid. More classifier examples would not be the first repair for this identified input/display mismatch.

Assess independent transfer

Selected case: Create two different images of width 4 and height 2, each with exactly three black pixels. Write both complete row encodings. Explain which fact their shared count retains and which difference it discards.

Selected case answer

One valid pair is A [1,1,1,0]/[0,0,0,0] and B [1,0,0,0]/[0,1,0,1]. Both have two rows of four values and three ones. Count retains the number of black positions but discards their locations. Other valid pairs are accepted if dimensions, counts and arrangements meet the constraints.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 6: Sound Becomes a Signal. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide large tactile or printed cells, with numbers as well as shading so colour is unnecessary. Let learners dictate row values. Extension: construct two grids with equal counts in every row but different arrangements, showing that row totals still lose information.

W4 is secure when both images are two by four, each has three ones, the patterns differ and the explanation identifies lost location information. If a learner draws valid pictures but mismatches their code, reteach the round-trip check, then assess with fresh patterns.

Week 6 transfers the idea of ordered numerical representations to sound, where the positions refer to times rather than image locations.

Plan the delayed return

Suggested return points in this level: ai-1/6, ai-1/8, ai-1/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-pixel count describes shape.